A CHARACTERISATION OF AFFINE FORMAL ALGEBRAIC SPACES BY THICKENINGS FROM AFFINE SCHEMES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The second condition of the statement is a set-theoretic condition recalled from a remark of the chapter rather than restated; that reference is carried as a navigation stub and the condition is recorded as part of the hypothesis.

Editorial note. The counted spans carry 2 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- An affine formal algebraic space admits a covering by thickenings and morphisms into it factor through a stage. Same chapter, Lemmas lemma-covering-by-thickenings and lemma-factor-through-thickening, invoked at proof lines 2108-2110. Give the easy direction of the characterisation.
- The definition of an affine formal algebraic space as a filtered colimit of affine schemes with thickenings as transition maps. Same chapter, Definition definition-affine-formal-algebraic-space, in force in the statement. Fixes what has to be produced in the converse.
- A representable thickening is a monomorphism of sheaves, and a morphism of schemes is a thickening precisely when it is a surjective closed immersion. Same chapter, the properties of thickenings used at proof lines 2122-2126. Show that the arrows of the constructed index category are thickenings.

Source reference. Original lemma 0AI9: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2057–2057. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AIA: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2080–2080. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Lemma 0.1. *Let S be a scheme. Let X be a sheaf on $(Sch/S)_{fppf}$. Then X is an affine formal algebraic space if and only if the following hold*

(1) *any morphism $U \to X$ where U is an affine scheme over S factors through a morphism $T \to X$ which is representable and a thickening with T an affine scheme over S, and*
(2) *a set theoretic condition as in Remark 0AIS.*

Proof. It follows from Lemmas 0AI9 and 0AIA that an affine formal algebraic space satisfies (1) and (2). In order to prove the converse we may assume X is not empty. Let Λ be the category of representable morphisms $T \to X$ which are thickenings where T is an affine scheme over S. This category is directed. Since X is not empty, Λ contains at least one object. If $T \to X$ and $T' \to X$ are in Λ, then we can factor $T \amalg T' \to X$ through $T'' \to X$ in Λ. Between any two objects of Λ there is a unique arrow or none. Thus Λ is a directed set and by assumption $X = \operatorname{colim}_{T \to X \text{ in } \Lambda} T$. To finish the proof we need to show that any arrow $T \to T'$ in Λ is a thickening. This is true because $T' \to X$ is a monomorphism of sheaves, so that $T = T \times_{T'} T' = T \times_X T'$ and hence the morphism $T \to T'$ equals the projection $T \times_X T' \to T'$ which is a thickening because $T \to X$ is a thickening. □